AoU

AXI-over-UCIe Bridge

Beta release | Open-source RTL | Apache 2.0

HIGHLIGHTS

- Carries AXI system-management and application traffic between chiplets
- Supports 256-, 512-, or 1024-bit AXI data per resource plane
- Adapts bus width so chiplets with different AXI data widths interoperate
- Provides one to four independently flow-controlled resource planes
- Handles UCIe FDI wake, clock, and state handshakes after activation
- Provides Advanced Peripheral Bus (APB) control, status, error reporting, and AXI aggregation
- Slang and Verilator lint targets integrated into the OCAH CI workflow

SYSTEM ROLE

AoU sits between the OCAH AXI fabric and an adopter-supplied Universal Chiplet Interconnect Express (UCIe) controller and PHY. It converts AXI transactions to and from the AoU packet protocol carried over the controller's Flit-aware die-to-die interface (FDI). OCAH supplies the protocol engine and bringup controller; the adopter supplies the UCIe controller, PHY, and physical link.

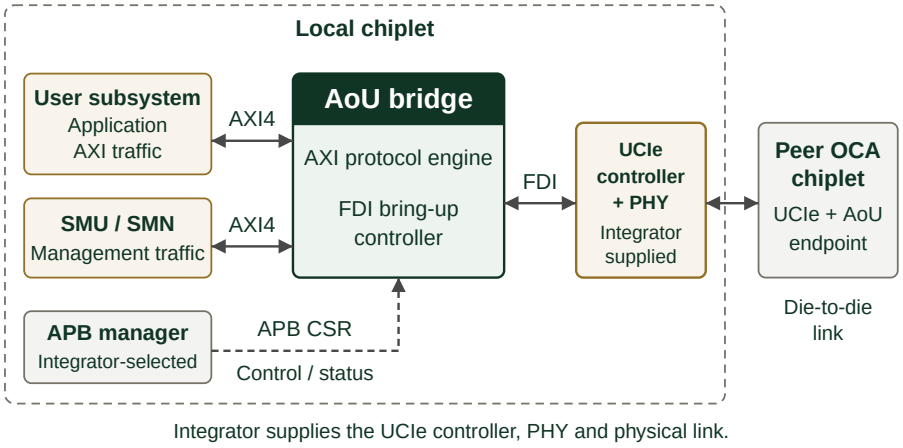
OVERVIEW

The AXI-over-UCIe Bridge (AoU) is OCAH's die-to-die transport component. Each of its one to four resource planes (RPs) presents a local AXI4 subordinate and a peer-facing AXI4 manager. After software requests activation, the controller performs the UCIe FDI handshakes and the engine exchanges AoU packets. Each plane adapts between its local AXI data width and the peer's, so chiplets with different bus widths interoperate.

The AoU RTL, developed jointly by Tenstorrent and BOS Semiconductors, provides the protocol engine and optional FDI bring-up controller. AoU attaches to a UCIe controller's FDI, not to a bare PHY; neither is a deliverable of this beta release, and the integrator supplies and qualifies both.

SYSTEM CONTEXT

AoU occupies the always-on domain alongside the System Management Unit (SMU) and System Management Network (SMN). The adopter owns the UCIe controller, PHY, and physical link; the peer need only be OCA-compliant.



The integrator connects the APB CSR port to a suitable manager.

AT A GLANCE

Feature	Description
AXI interface (per RP)	512-bit data (reference; 256 or 1024 selectable); 64-bit address; 10-bit ID
Outstanding transactions	32 reads and 32 writes per manager/subordinate path
FDI	Single-PHY 32 B / 256-bit data (reference); up to 128 B / 1024-bit data
APB CSR / IP version	32-bit address and data / major 1, minor 0

The reference uses one resource plane, 512-bit AXI data, and `FDI_CFG_SP_32B`. Aggregation and low-power controls are runtime-configurable; FIFO depths are build-time parameters.

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CAPABILITIES

Protocol bridge

- Converts AXI4 read and write transactions to AoU packet protocol and back; supports INCR bursts (FIXED is not supported), split transactions, and write-response early-return mode
- Per-plane upsizer and downsizer adapt the local AXI data width to the peer's, so attached chiplets need not share a bus width
- Per-plane AXI aggregator packs narrow beats into full-width local AXI beats to improve local bus utilization (runtime-configurable)
- Request and response credit streams managed independently during link deactivation, enabling in-flight responses to complete before reactivation

Link management

- Integrated UCIe FDI bringup controller handles wake, clock, state, and receive-active handshake sequences after software requests activation
- Software-initiated link activation and deactivation via APB registers; status bits report FSM state and transaction completion
- Configurable deactivation timeout and low-power transmit mode with tunable heartbeat threshold

Observability and error handling

- Per-plane error information registers capture AXI response errors and ID-mismatch events
- Maskable interrupts for link-reset events (activation/deactivation ACK timeout, invalid activation message, message-credit timeout), early write-response errors, and master/slave ID mismatches
- Software-controlled reset of the AoU core logic

INTERFACES AND CONFIGURATION

Feature	Integration options
Clocks and resets	Core and APB clock/reset domains
AXI subordinate (per RP)	AXI4 subordinate accepting traffic from the local fabric
AXI manager (per RP)	AXI4 master issuing transactions toward peer chiplet
FDI data plane	To the UCIe controller; PHY0 always present, PHY1 with `+define+TWO_PHY`
APB CSR	32-bit APB subordinate for link control, status, and error registers
Build-time	`RP_COUNT` (1-4), `FDI_CONFIG`, per-plane `AXI_DATA_WD`, FIFO depths, outstanding counts

`FDI_CONFIG` selects the single-PHY width or two-PHY width pair using one of five values: `FDI_CFG_SP_32B`, `FDI_CFG_SP_64B`, `FDI_CFG_SP_128B`, `FDI_CFG_TP_32B_64B`, `FDI_CFG_TP_64B_128B`). Two-PHY builds require `+define+TWO_PHY`; selecting a `FDI_CFG_TP_*` value without this define (or vice versa) is the integrator's responsibility to prevent.

INTEGRATION DEPENDENCIES

- Supply and qualify the UCIe controller, PHY, and physical link; AoU cannot drive a bare PHY. Keep their power domain stable before FDI bringup.
- Match `FDI_CONFIG` to `+define+TWO_PHY`; RTL does not check this combination.
- Connect the AXI ports and Advanced Peripheral Bus (APB) CSR subordinate to the product fabrics, and define address decoding and peer routing.
- Account for asynchronous core/APB clocks using the included bridge, and activate the link in software before issuing AXI traffic.

VERIFICATION AND MATURITY

The vendored package passes the OCAH Slang and Verilator lint targets. The current OCAH SMU RTL and simulation environment do not instantiate AoU, and no OCAH-specific functional simulation, formal verification, or coverage results are claimed. Upstream verification is described in the AoU TRM.

No throughput, latency, area, power, frequency, or silicon-characterization results are claimed.

DELIVERABLES

- SystemVerilog protocol-engine and FDI controller RTL with dependency filelists
- SystemRDL CSR source and generated register RTL
- OCAH lint and synthesis integration collateral
- Apache License 2.0 repository distribution

RESOURCES

Documentation: OCAH documentation
Protocol: AoU TRM
RTL and registers: `vendor/tenstorrent/aou-rtl/upstream/`
OCAH flow: `vendor/tenstorrent/aou-rtl/overlay/flow.mk`

TERMS

AoU: AXI-over-UCIe bridge. APB: Advanced Peripheral Bus. FDI: Flit-aware die-to-die interface. RP: resource plane. SMN: System Management Network. SMU: System Management Unit.

DOCUMENT CONTROL

Version	Date	Author	Change